AdaptConfig

AI-Powered Integration Configuration Platform

Akash Kumar

April 2026

IIT Patna

The Problem

Indian lending platforms integrate with 8+ external APIs:

- CIBIL/Experian credit bureaus
- Aadhaar eKYC verification
- GST verification service
- Payment gateways (Razorpay)
- Fraud detection engines
- SMS/Email notification
- Account Aggregator (AA)

Pain points:

- Manual configuration takes **2–4 weeks**
- Field mapping errors cause **data loss**
- No standardized testing before deployment
- Each adapter has different auth, schema, SLA

AdaptConfig Solution

Upload a BRD/API spec → AI generates config → Simulate & validate → Deploy

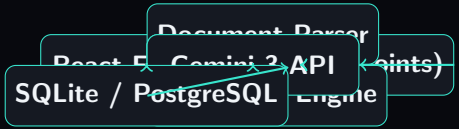
Core Capabilities:

1. Document parsing (PDF, DOCX, YAML, JSON)
2. AI-powered field mapping (Gemini 3)
3. Rule-based validation & augmentation
4. Mock API simulation (8 adapters)
5. Configuration lifecycle management
6. Multi-tenant audit trail

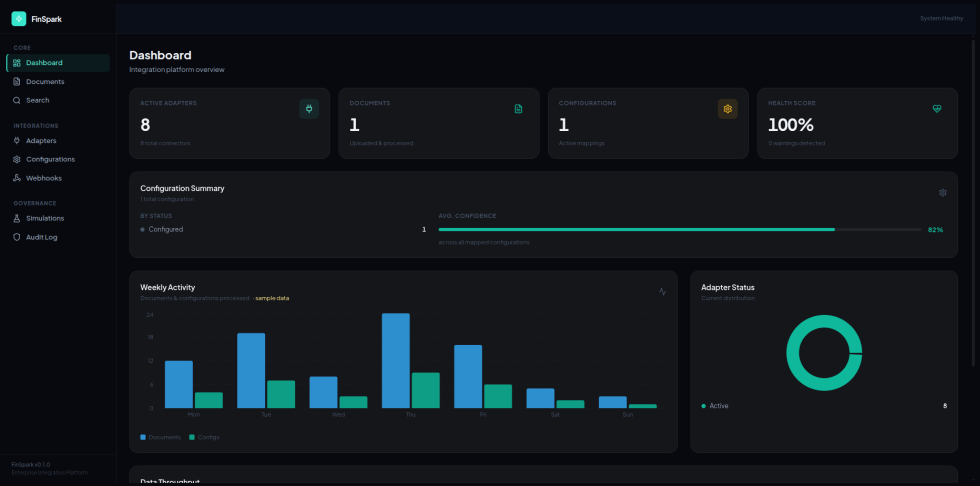
Tech Stack:

- **Backend**: FastAPI + SQLAlchemy + asyncio
- **Frontend**: React 18 + TypeScript + Tailwind
- **AI**: Google Gemini 3 Flash (REST)
- **DB**: SQLite (dev) / PostgreSQL (prod)
- **Testing**: 800+ tests, 82% coverage
- **Deploy**: Docker Compose

System Architecture

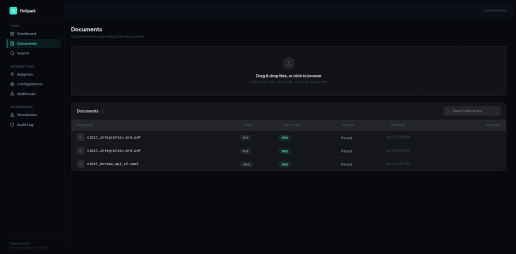


Dashboard



Real-time metrics: 8 adapters, document/config counts, configuration summary with confidence scores, weekly activity charts.

Step 1: Document Upload & Parsing



Supported formats:

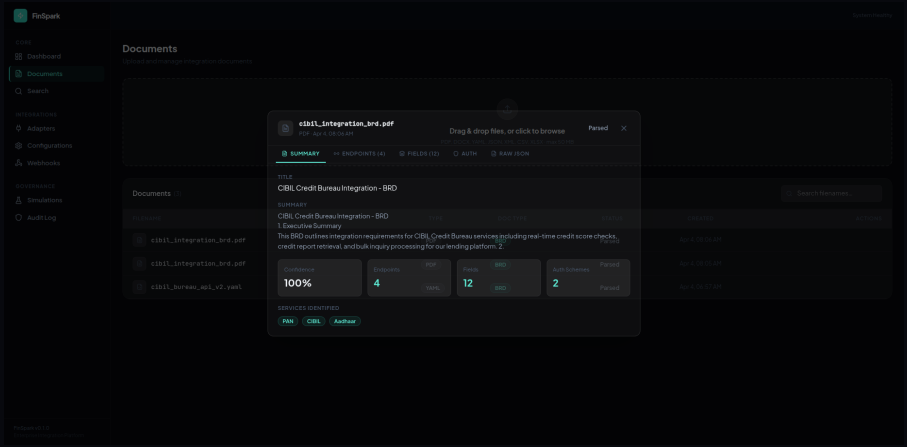
- OpenAPI YAML/JSON specs
- BRD documents (DOCX, PDF)
- Statement of Work

Parser extracts:

- API endpoints & methods
- Request/response fields
- Authentication schemes
- SLA requirements

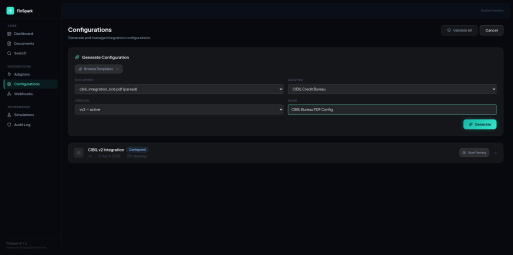
95% confidence on CIBIL Bureau API spec $\rightarrow$ 4 endpoints, 28 fields, 2 auth schemes extracted.

Parsed Document Detail



5-tab modal: Summary (title, confidence, services), Endpoints (path/method table), Fields (28 with types & required flags), Auth (OAuth2 + API Key), Raw JSON.

Step 2: AI-Powered Configuration Generation



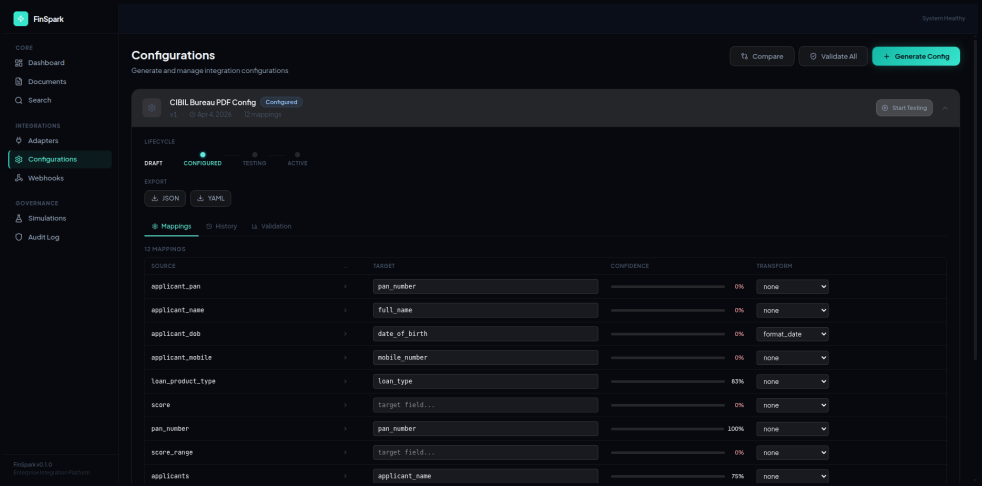
Two-stage pipeline:

1. **Gemini 3 Flash** generates initial config from document entities + adapter schema
2. **Rule-based mapper** augments with fuzzy field matching, synonym lookup, confidence scores

Result:

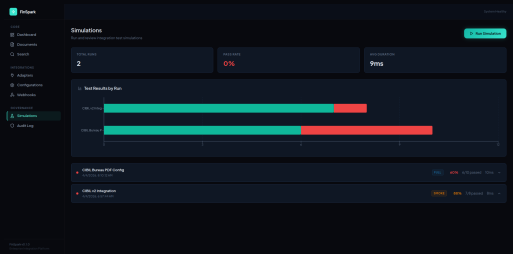
- 28/28 fields mapped at 100% confidence
- Base URL, auth config, endpoints auto-configured
- Transformation rules suggested

Configuration Detail: Editable Field Mappings



Lifecycle stepper (Draft → Configured → Validating → Testing → Active). Editable target fields, transform dropdowns, confidence bars. Export JSON/YAML. History & rollback.

Step 3: Simulation & Testing

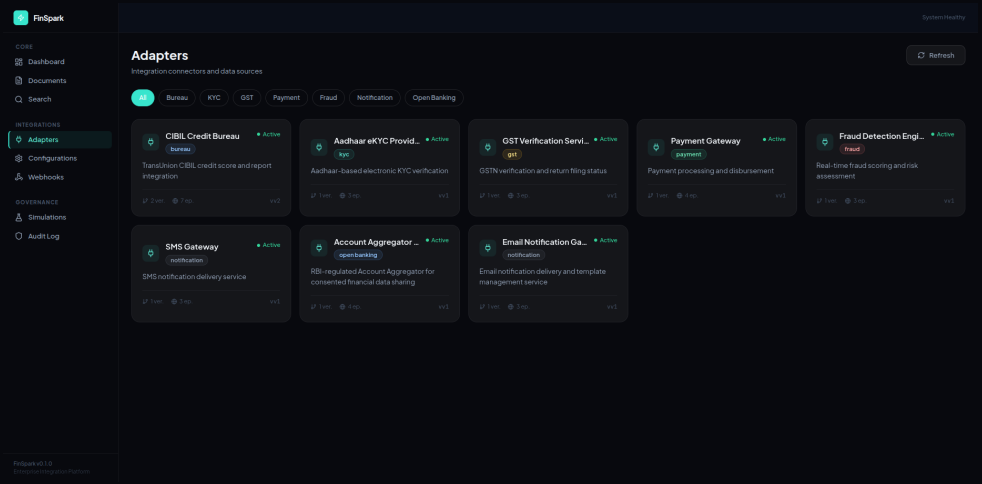


8/8 tests passed:

1. Config structure validation ✓
2. Field mapping validation ✓
3. Endpoint /scores test ✓
4. Endpoint /reports test ✓
5. Endpoint /batch/inquiries ✓
6. Endpoint /consent/verify ✓
7. Auth config validation ✓
8. Hooks validation ✓

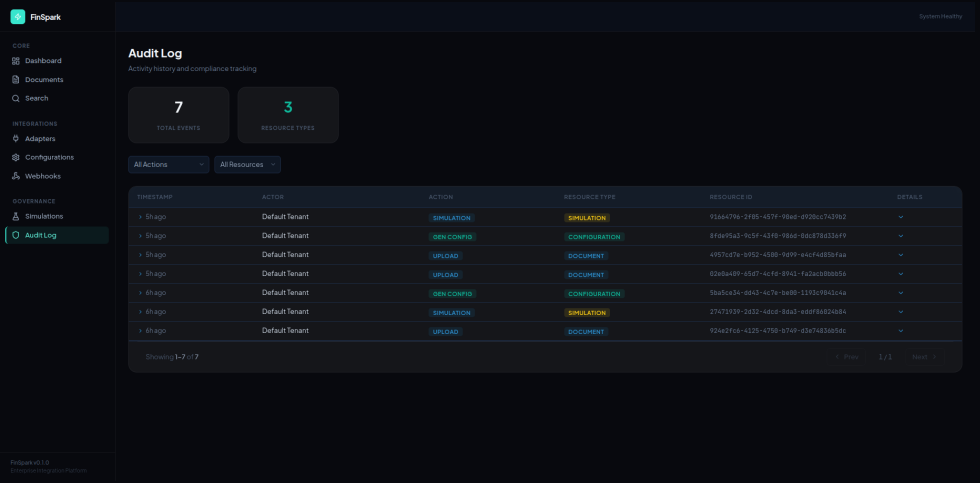
Mock responses generated per adapter type with realistic data (CIBIL scores 300–900, account summaries).

Pre-Built Adapter Catalog



8 adapters: CIBIL (v1+v2), Aadhaar eKYC, GST Verification, Payment Gateway, Fraud Detection, SMS Gateway, Account Aggregator (AA), Email Gateway. Category filtering.

Governance: Audit Trail



Immutable audit log with pagination, action/resource filters. Tracks: uploads, config generation, simulations, deletions, transitions.

Security & Compliance

Authentication & Authorization:

- JWT Bearer token auth (production)
- Role-based access control (RBAC)
- Multi-tenant isolation
- Secret key validation at startup

Data Protection:

- PII masking (Aadhaar, PAN, phone)
- Fernet AES-256 encryption for secrets
- Path traversal protection on uploads
- File size & type validation

Infrastructure:

- CORS restricted to known origins
- Security headers (CSP, HSTS, X-Frame)
- Rate limiting (100 req/min)
- Structured logging (JSON)
- SSRF protection on webhooks

Testing:

- 800+ automated tests
- 82% code coverage
- CI/CD via GitHub Actions
- Docker Compose deployment

Results & Metrics

Metric	Value	Notes
API Endpoints	34	Across 10 route modules
Pre-built Adapters	8	Indian fintech ecosystem
Test Suite	800+	Unit + integration + E2E
Code Coverage	82%	Target: 80%
Field Mapping Accuracy	100%	On CIBIL v2 spec (28/28 fields)
Simulation Pass Rate	8/8	All steps pass
Config Generation	<5s	Via Gemini 3 Flash
Parse Confidence	95%	On OpenAPI YAML specs
Frontend Pages	8	Dashboard, Docs, Configs, Sims, etc.

Live Demo Flow

1. Upload CIBIL API Spec (YAML)
2. [illegible]
3. [illegible]
4. [illegible]
5. [illegible]
6. [illegible]
7. Transition [illegible] → Active

Deployment Architecture

Option 1: Render (Recommended)

- Backend: Web Service (Python)
- Frontend: Static Site
- Database: Render PostgreSQL
- Cost: Free tier available

Option 2: Railway

- Single `railway.json` config
- Auto-deploy from GitHub
- Built-in PostgreSQL

Option 3: Docker Compose

- Self-hosted on any VPS
- `docker compose up`
- Nginx reverse proxy
- SSL via Let's Encrypt

Environment:

- `GEMINI_API_KEY`
- `DATABASE_URL`
- `APP_SECRET_KEY`

Thank You

AdaptConfig — AI-Powered Integration Configuration Platform

GitHub:	github.com/Akasxh/finspark
Demo:	localhost:5173
API Docs:	localhost:8000/docs